

Patent Application

For

**Adaptive Cognitive Data Fabric with Dynamic
Pipeline Orchestration and Immutable
Federated Governance**

APPLICANTS/INVENTORS

1. Santhosh Kumar PENDYALA, Address: Mckinney, TX, 75070 USA
2. Vijay Kumar ADARI, Address: Prosper, TX, 75078 USA
3. Satyanarayana Murthy POLISETTY, Address: Raleigh, North Carolina, 27616 USA
4. Sushil Prabhu PRABHAKARAN, Address: Prosper, TX 75078 USA
5. Rakesh MITTAPALLY, Address: Aldie, Virginia, 20105 USA
6. Surya Rao RAYARAO, Address: Overland Park, KS, 66085 USA
7. Nitesh RAMANCHA, Address: Mckinney, TX, 75071 USA
8. Satyanarayana BALLAMUDI, Address: Celina, TX, 75009 USA
9. Mahesh Kumar MISHRA, Address: McKinney, TX, 75072 USA
10. Chandra Shekar CHENNAMSETTY, Address: Antioch, CA 94531 USA
11. Nagababu KANDULA, Address: Liberty township, Ohio, 45044 USA

Adaptive Cognitive Data Fabric with Dynamic Pipeline Orchestration and Immutable Federated Governance

BACKGROUND OF THE INVENTION

FIELD OF THE INVENTION

[0001] The present invention relates to intelligent data management, federated governance, and secure AI-driven processing within complex data ecosystems. More specifically, it pertains to a self-evolving adaptive cognitive data fabric system that integrates quantum-resistant security, AI-driven pipeline orchestration, privacy-preserving semantic profiling, and federated metadata governance to autonomously manage heterogeneous data streams, optimize computational workloads, and enforce regulatory compliance across distributed environments.

DESCRIPTION OF THE RELATED ART

[0002] The rapid expansion of multi-source, multi-modal data across enterprises has introduced significant challenges in data management, governance, and compliance. Traditional data management systems are often rigid, requiring extensive manual intervention for error correction, workflow optimization, and governance enforcement. These legacy systems struggle with real-time adaptability, fail to integrate heterogeneous data sources efficiently, and lack robust security against emerging threats such as quantum-based cryptographic attacks.

[0003] Existing federated data systems lack mechanisms to balance privacy, security, and computational efficiency while preserving model transparency and lineage tracking. AI model training across distributed environments is often constrained by privacy regulations, leading to data silos that limit collaborative analytics. Furthermore, conventional metadata management approaches lack immutability and verifiable compliance tracking, making them susceptible to tampering and regulatory inconsistencies.

[0004] As enterprise data systems continue to scale across hybrid cloud, edge, and multi-cloud environments, the demand for an intelligent, self-optimizing, and regulation-aware data fabric has become critical. A system that integrates federated governance, self-healing

orchestration, and adaptive AI-driven optimization is necessary to enable seamless, secure, and automated data management.

SUMMARY OF THE INVENTION

[0005] The Self-Evolving Adaptive Cognitive Data Fabric System is an intelligent and resilient data management framework designed to autonomously optimize data workflows, enforce federated governance, and enhance AI-driven processing. The system introduces novel innovations in secure data ingestion, self-healing pipeline orchestration, and privacy-preserving semantic profiling to enable scalable and regulation-compliant data operations.

[0006] At its core, the system includes a quantum-resistant data collector that ensures secure, multi-protocol ingestion with post-quantum cryptographic safeguards. The self-healing stream reconciliation engine utilizes Byzantine fault tolerance to maintain synchronization and consistency across heterogeneous data streams. The real-time semantic profiler applies natural language processing and computer vision to extract insights from multi-modal data while ensuring privacy through differential privacy techniques.

[0007] A key invention of the system is the adaptive pipeline orchestrator, which employs a multi-agent reinforcement learning framework to dynamically optimize data workflows. The orchestrator leverages hierarchical reward mechanisms to continuously refine task execution, resource allocation, and model training cycles. Complementing this capability, the cognitive model re-trainer integrates continual learning and curriculum-based knowledge distillation to mitigate model drift and ensure sustained predictive accuracy.

[0008] The system also features a federated metadata manager that enforces immutable provenance tracking using blockchain technology with zero-knowledge proofs, ensuring compliance with evolving regulatory requirements. Additionally, an AI-driven policy enforcement engine translates natural language governance rules into formally verified, rule-based implementations, enhancing transparency and security.

[0009] By integrating these advanced capabilities, the invention provides an end-to-end autonomous data fabric that optimizes AI-driven analytics, ensures cross-jurisdictional regulatory adherence, and enables seamless federated collaboration across distributed data ecosystems. This self-evolving, adaptive, and privacy-aware system redefines enterprise data

governance and intelligence processing for the next generation of scalable, secure, and resilient data architectures.

BRIEF SUMMARY

[0010] Embodiments of the present disclosure may include a self-evolving adaptive cognitive data fabric system including a quantum-resistant data collector configured for secure, multi-protocol ingestion with dynamic key rotation and post-quantum cryptographic algorithms. Embodiments may also include a stream reconciliation engine that implements self-healing consistency with Byzantine fault tolerance to reliably synchronize heterogeneous data streams.

[0011] Embodiments may also include a real-time semantic profiler that utilizes differential privacy-preserving vector embedding generation by integrating natural language processing and computer vision for multi-modal data analysis. Embodiments may also include an adaptive pipeline orchestrator employing a multi-agent reinforcement learning framework with a hierarchical reward system and automatic feature engineering based on semantic feedback. Embodiments may also include a cognitive model re-trainer configured for incremental knowledge distillation with continual learning and curriculum-based retraining to address concept drift. Embodiments may also include a federated metadata manager that enforces immutable, blockchain-based provenance tracking with zero-knowledge proofs.

[0012] Embodiments of the present disclosure may also include a method for dynamic optimization of data processing workflows in an adaptive cognitive data fabric, including continuously ingesting and reconciling heterogeneous data streams via a quantum-resistant data collector and self-healing reconciliation engine. Embodiments may also include analyzing the ingested data through a differential privacy-preserving semantic profiler to extract schema, content, and multi-modal semantic meaning.

[0013] Embodiments may also include dynamically reconfiguring processing workflows using an adaptive pipeline orchestrator that leverages multi-agent reinforcement learning with a hierarchical reward mechanism and automatic feature engineering. Embodiments may also include triggering retraining of cognitive models through incremental knowledge distillation and curriculum-based prioritization upon detecting performance drift. Embodiments may also include maintaining immutable metadata and provenance records via a federated governance framework employing blockchain-based verification and zero-knowledge proofs.

[0014] Embodiments of the present disclosure may also include a self-adaptive federated governance and policy enforcement system for a cognitive data fabric, including a federated metadata manager that maintains a centralized yet distributed repository of data catalogs with blockchain-based immutable tracking and zero-knowledge proofs. Embodiments may also include a policy enforcement engine that translates natural language policies into formally verified, rule-based implementations and utilizes homomorphic encryption for real-time access control.

[0015] Embodiments may also include an explainability service that generates counterfactual explanations through causal model extraction to enhance decision transparency. Embodiments may also include a resource optimization module integrating a market-based predictive resource allocator, dynamic scaling engine, topology-aware peer discovery for secure P2P mesh communication, and carbon-aware scheduling to optimize computing resources across heterogeneous environments. Embodiments may also include dependent Claims:.

[0016] Embodiments of the present disclosure may also include, the system of Final Independent Claim 1. In some embodiments, the adaptive pipeline orchestrator employs a multi-agent reinforcement learning framework with a hierarchical reward system balancing immediate processing efficiency, computational constraints, and long-term learning objectives.

[0017] Embodiments of the present disclosure may also include, the system of Final Independent Claim 2. In some embodiments, the real-time semantic profiler implements multi-modal vector embedding alignment that dynamically adjusts embedding dimensions based on semantic similarity metrics to maintain a unified representation across diverse data types.

[0018] Embodiments of the present disclosure may also include, the system of Final Independent Claim 1. In some embodiments, the cognitive model re-trainer implements curriculum learning to prioritize retraining on samples that most effectively address performance gaps while maintaining a directed acyclic graph (DAG) of model derivations for automated performance comparison.

[0019] Embodiments of the present disclosure may also include, the system of Final Independent Claim 3. In some embodiments, the explainability service generates counterfactual explanations using an invertible transformation map between input features and model decisions to enable bidirectional traceability of decision pathways.

[0020] Embodiments of the present disclosure may also include, the system of Final Independent Claim 1. In some embodiments, the federated metadata manager implements cross-jurisdictional compliance verification using selective disclosure protocols integrated with blockchain-based immutable record-keeping.

[0021] Embodiments of the present disclosure may also include, the system of Final Independent Claim 3. In some embodiments, the topology-aware peer discovery system implements a self-organizing compute mesh that dynamically redistributes workloads based on network conditions, hardware capabilities, and energy efficiency metrics.

[0022] Embodiments of the present disclosure may also include, the system of Final Independent Claim 1. In some embodiments, the quantum-resistant data collector incorporates forward secrecy protocols with post-quantum cryptographic primitives to ensure robust security against future quantum computing attacks.

[0023] Embodiments of the present disclosure may also include, the system of Final Independent Claim 3. In some embodiments, the natural language policy compiler automatically translates high-level governance policies into formally verified, rule-based enforcement mechanisms with mathematical guarantees of compliance across diverse technology stacks.

BRIEF DESCRIPTION OF THE FIGURES

[0024] FIG. 1A is a block diagram illustrating a Self-Evolving Adaptive Cognitive Data Fabric System, according to some embodiments of the present disclosure.

[0025] FIG. 1 is a System Architecture for Advanced Cognitive Data Fabric with Federated Governance and Dynamic workflows Optimization, according to some embodiments of the present disclosure.

[0026] FIG. 2A is a flowchart illustrating a method for Dynamic data processing workflows in an Adaptive Cognitive Data Fabric, according to some embodiments of the present disclosure.

[0027] FIG. 2 is a method diagram of Advanced Cognitive Data Framework Methodology, according to some embodiments of the present disclosure.

[0028] FIG. 3 is a method diagram for Cognitive Intelligence Layer and Pipeline Execution Mesh system, according to some embodiments of the present disclosure.

[0029] FIG. 4 is a method diagram for Multi-modal Semantic Intelligence Processing Fabric system, according to some embodiments of the present disclosure.

[0030] FIG. 5A is block diagram of Self-Adaptive Federated Governance and Policy Enforcement system, according to some embodiments of the present disclosure.

[0031] FIG. 6 is method diagram of Adaptive Cognitive Data Fabric - Federated Data Governance and Cognitive Model Management system, according to some embodiments of the present disclosure.

[0032] FIG. 7 is a block diagram System and Method for Dependent Claims Detailed Description system, according to some embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE FIGURES & INVENTION

FIG. 1A Self-Evolving Adaptive Cognitive Data Fabric System Block Diagram

FIG. 1 is a block diagram that describes a self-evolving adaptive cognitive data fabric system 100, according to some embodiments of the present disclosure. In some embodiments, the self-evolving adaptive cognitive data fabric system 100 may include a stream reconciliation engine 112 that implements self-healing consistency with Byzantine fault tolerance to reliably synchronize heterogeneous data streams, a real-time semantic profiler 113 that utilizes differential privacy-preserving vector embedding generation by integrating natural language processing and computer vision for multi-modal data analysis, an adaptive pipeline orchestrator 115 employing a multi-agent reinforcement learning framework with a hierarchical reward system and automatic feature engineering based on semantic feedback, and a cognitive model re-trainer 116 configured for incremental knowledge distillation with continual learning and curriculum-based retraining to address concept drift.

[0033] In some embodiments, the self-evolving adaptive cognitive data fabric system 100 may also include a quantum-resistant data collector 111 configured for secure, multi-protocol ingestion with dynamic key rotation and post-quantum cryptographic algorithms. The self-evolving adaptive cognitive data fabric system 100 may also include a federated metadata

manager 104 that enforces immutable, blockchain-based provenance tracking with zero-knowledge proofs.

[0034] The Quantum-Resistant Data Collector 111 is a highly secure data ingestion module designed to facilitate the transmission of structured and unstructured data across multiple trust zones while ensuring resistance against quantum-based cryptographic attacks. This component employs a quantum-resistant encryption scheme with dynamic key rotation to protect data as it traverses distributed nodes, mitigating risks associated with future cryptanalytic advancements. The data collector operates in conjunction with the Stream Reconciliation Engine to validate data consistency before passing it to processing pipelines. It supports homomorphic encryption to enable secure computation on encrypted datasets, ensuring compliance with privacy regulations. This module interacts with the Federated Metadata Manager to record data provenance and maintain immutable metadata tracking, ensuring end-to-end visibility of data sources and transformations within the system.

[0035] The Stream Reconciliation Engine 112 is a dynamic processing module that ensures data consistency and synchronization across multiple real-time data streams. It achieves this by employing sophisticated algorithms to detect discrepancies, apply reconciliation rules, and resolve conflicts in distributed data environments. The Stream Reconciliation Engine continuously ingests data from various sources, utilizing the Reconciliation Rules Engine to assess data integrity and match records. In case of discrepancies, the Conflict Resolution Unit intelligently applies predefined resolution strategies to maintain stream consistency. This engine supports real-time synchronization through the Stream Synchronization Layer, ensuring data alignment and minimizing delays. Furthermore, the Stream Reconciliation Engine provides telemetry data to the Adaptive Pipeline Orchestrator, enabling real-time adjustments based on performance metrics and resource availability. Through these capabilities, the Stream Reconciliation Engine ensures the smooth and accurate processing of data across disparate sources.

[0036] The Real-Time Semantic Profiler 113 is a core analytical module responsible for extracting schema structures, contextual insights, and semantic embeddings from ingested data streams. This component applies a combination of natural language processing, computer vision, and structured data analysis techniques to generate enriched representations of data, enabling downstream components to make context-aware decisions. The semantic profiling process employs differential privacy-preserving techniques to ensure that sensitive data

attributes remain protected while still contributing to analytical workflows. The profiler interfaces with the Adaptive Pipeline Orchestrator to provide semantic metadata that guides pipeline configuration, ensuring that data transformations and processing tasks align with the intrinsic characteristics of the data. Additionally, it logs metadata into the Federated Metadata Manager, contributing to comprehensive data lineage tracking and governance compliance.

[0037] The Adaptive Pipeline Orchestrator 115 is a self-optimizing workflow management engine that dynamically configures and executes data processing pipelines based on contextual information and performance objectives. This module employs a multi-agent reinforcement learning framework with a hierarchical reward system to optimize pipeline efficiency across multiple objectives, including latency, computational cost, and data quality. By continuously monitoring pipeline execution metrics, the Adaptive Pipeline Orchestrator refines task scheduling and resource allocation strategies in real-time. The orchestrator integrates with the Federated Metadata Manager to update pipeline metadata and provenance records, ensuring traceability and compliance. It also collaborates with the Heterogeneous Inference Accelerator to allocate inference workloads efficiently, enhancing end-to-end processing performance across cloud and edge environments.

[0038] The Cognitive Model Re-trainer 116 is an advanced machine learning lifecycle management module designed to ensure continuous model adaptation in response to evolving data distributions. This module employs incremental knowledge distillation techniques and curriculum learning strategies to enable models to retain prior knowledge while incorporating new insights from retraining cycles. By utilizing multivariate statistical drift detection, the Cognitive Model Re-trainer identifies performance degradation in deployed models and triggers targeted retraining processes. It maintains a directed acyclic graph (DAG)-based versioning system to track model lineage and rollback capabilities, ensuring that prior model versions remain accessible for reproducibility and compliance audits. Additionally, this module collaborates with the Adaptive Pipeline Orchestrator to optimize retraining workflows, reducing computational overhead while maintaining high predictive accuracy across evolving datasets.

[0039] The Federated Metadata Manager 104 serves as the cornerstone of metadata governance by maintaining immutable records of metadata provenance, lineage, and policy enforcement across distributed environments. This component leverages blockchain-based tracking mechanisms to ensure tamper-proof auditability of metadata changes, while

employing zero-knowledge proofs for selective disclosure of metadata attributes. By integrating cryptographic verification techniques, the Federated Metadata Manager enforces fine-grained access control policies, ensuring that only authorized entities can modify or access metadata records. It synchronizes with cloud and edge storage environments to maintain metadata consistency across diverse infrastructures, thereby preventing fragmentation of metadata governance. Additionally, it collaborates with the Policy Enforcement Engine to translate high-level governance policies into executable rules that govern metadata access, storage, and utilization.

[0040] The Heterogeneous Inference Accelerator 117 is a computational optimization module that dynamically partitions machine learning inference workloads across mixed-precision hardware environments, including CPUs, GPUs, TPUs, and specialized accelerators. This module employs adaptive batching techniques to optimize inference throughput while minimizing latency across distributed inference clusters. By leveraging workload profiling and hardware-aware scheduling, the Heterogeneous Inference Accelerator ensures that compute-intensive deep learning models execute efficiently on available hardware resources. The module integrates with the Cognitive Model Re-trainer to apply transfer learning techniques, enabling efficient model adaptation across different hardware configurations. Additionally, it provides execution telemetry to the Adaptive Pipeline Orchestrator, allowing real-time adjustments to inference workloads based on resource availability and performance constraints.

[0041] The Multimodal Intelligence Processing Fabric 416 is a high-performance analytical framework designed to unify and process heterogeneous data modalities, including structured text, images, audio, and video. This module incorporates domain-specific feature extraction techniques, leveraging deep learning and statistical modeling to generate semantically meaningful representations across modalities. The fabric applies cross-modal fusion strategies to enhance contextual understanding, enabling downstream AI models to make holistic inferences from diverse data sources. The processing fabric interacts with the Real-Time Semantic Profiler to refine data representations and improve feature alignment between modalities. It also integrates with the Cognitive Model Re-trainer to support continuous learning across multimodal datasets, ensuring that inference models remain robust in dynamically evolving environments. Additionally, it synchronizes with the Adaptive Pipeline Orchestrator to enable seamless execution of multimodal processing workflows, enhancing the system's overall adaptability and intelligence.

[0042] The Holistic Governance Repository 509 is a centralized framework for managing and enforcing data governance policies across an organization's data ecosystem. This repository consolidates all governance-related metadata, including compliance regulations, data stewardship guidelines, and auditing protocols, in the Governance Policy Storage. It employs the Metadata Management Layer to organize and categorize this information for easy access and analysis. The Policy Enforcement Engine actively applies governance rules to ensure that all data handling practices adhere to regulatory standards and internal policies. The Audit & Compliance Module tracks all policy enforcement activities, ensuring that data governance processes are fully auditable and compliant with industry regulations. The Access Control Layer secures access to sensitive governance data, ensuring proper authorization for stakeholders based on their roles. By integrating these components, the Holistic Governance Repository provides a robust governance infrastructure that supports transparent, secure, and compliant data management across the organization.

[0043] The Policy Enforcement Engine 105 is responsible for dynamically translating abstract governance policies into machine-executable enforcement rules, ensuring compliance with data protection regulations and internal security protocols. This module integrates a natural language policy compiler with formal verification mechanisms to guarantee that all policy enforcement actions adhere to predefined constraints. By utilizing rule-based and AI-driven approaches, it continuously refines enforcement strategies based on evolving compliance requirements. The Policy Enforcement Engine works in conjunction with the Real-Time Access Control module to manage data access permissions while maintaining audit logs of policy execution. Furthermore, it interfaces with the Explainability Service to provide transparent justifications for policy decisions, thereby improving regulatory compliance and fostering trust in automated governance mechanisms.

[0044] Through these methods, the Adaptive Cognitive Data Fabric (ACDF) establishes a integration of federated metadata governance, quantum-resistant security, real-time semantic profiling, and multimodal inference acceleration enables enterprises to optimize their AI-driven workflows while maintaining compliance and operational resilience. The system's ability to dynamically reconfigure pipelines, adapt machine learning models, and enforce governance policies positions it as a scalable and intelligent data processing framework for enterprise applications.

FIG. 1 System Architecture for Advanced Cognitive Data Fabric with Federated Governance and Dynamic workflows Optimization

FIG. 1 illustrates the architecture diagram of an Advanced Cognitive Data Fabric (ACDF) system, according to embodiments of the present disclosure. This system introduces a comprehensive framework for managing heterogeneous data across distributed environments, integrating state-of-the-art security, adaptive processing, and robust governance mechanisms with multiple patentable innovations that ensure dynamic scalability and resilience.

[0045] The Enhanced ACDF System Architecture is organized into five principal layers: Data Sources 118, Data Ingestion 119, Processing Modules 120, Governance & Management 107, and Distributed Compute Environment 108. Each layer comprises specialized components that work synergistically to provide secure data acquisition, intelligent processing, and efficient resource management. This design is uniquely engineered to address challenges associated with processing heterogeneous data types, ensuring that both structured and unstructured information are processed with optimal accuracy and security.

[0046] The Data Sources layer 118 forms the foundation of the system by sourcing data from diverse origins. It is divided into two main categories. The Structured Data component 109 includes traditional time-series databases, relational databases, graph databases, and data warehouses, which supply well-organized, schema-defined data essential for high-integrity analytics. The Unstructured Data component 110 handles more amorphous inputs such as text documents, images, audio/video streams, and sensor telemetry, all of which require advanced processing techniques to extract actionable insights.

[0047] The Data Ingestion Layer 119 is designed to securely acquire and reconcile data from multiple sources, ensuring that information flows reliably into the system. The Quantum-Resistant Data Collector 111 is a key component that implements dynamic key rotation using quantum-resistant encryption schemes to secure data transmission across various trust zones. Complementing this, the Stream Reconciliation Engine 112 employs a self-healing consistency mechanism with Byzantine fault tolerance, effectively maintaining the integrity of distributed data streams even under adverse network conditions. These innovations provide a robust front line of defense against emerging security threats and operational disruptions.

[0048] At the heart of the system, the Processing Modules 120 execute advanced data transformation and intelligent decision-making processes. The Real-Time Semantic Profiler

113 generates semantic vector embeddings in real time while incorporating differential privacy-preserving techniques with a dynamic privacy budget allocation, ensuring that sensitive information remains secure during analysis. The Adaptive Pipeline Orchestrator 115 dynamically manages and optimizes workflow execution through context-aware resource allocation, leveraging multi-objective optimization to meet fluctuating processing demands. Meanwhile, the Cognitive Model Re-trainer 116 continuously refines analytical models using incremental knowledge distillation and concept drift detection to sustain high performance over time. The Heterogeneous Inference Accelerator 117 further enhances efficiency by dynamically partitioning models across mixed precision hardware and employing adaptive batching strategies, thereby accelerating inference tasks while optimizing resource utilization.

[0049] The Governance & Management layer 107 provides comprehensive oversight and control to ensure system compliance, traceability, and integrity. The Federated Metadata Manager 104 utilizes blockchain-based immutable provenance tracking coupled with zero-knowledge proofs to manage metadata securely and support selective disclosure of critical information. The Policy Enforcement Engine 105 incorporates a natural language policy compiler with formal verification processes, ensuring that all governance rules are applied consistently throughout the system. Additionally, the Explainability Service 106 enhances system transparency by generating counterfactual explanations through causal model extraction, which aids in understanding and validating the decision-making processes.

[0050] The Distributed Compute Environment 108 facilitates scalable and resilient processing by seamlessly integrating both edge and cloud resources. The Edge Node Cluster 101 is engineered to support latency-sensitive applications by utilizing topology-aware peer discovery and predictive compute offloading, thus optimizing processing at the network edge. The Cloud Compute Nodes 102 offer elastic resource orchestration with workload-specific hardware reconfiguration, ensuring that computational capacity scales dynamically to meet demand. The Hybrid Storage Manager 103 underpins the environment by implementing auto-tiering data placement and access pattern-based caching, thereby enhancing storage performance and reducing overall costs through intelligent data management across distributed storage tiers.

[0051] Emphasis is placed on the advanced components that drive the system's patentable innovations. The Quantum-Resistant Data Collector and Stream Reconciliation Engine in the Data Ingestion Layer form a robust security perimeter, while the Processing Modules—

especially the Adaptive Pipeline Orchestrator and Cognitive Model Re-trainer are designed to ensure continuous, adaptive learning and efficient workflow management. In the Governance & Management layer, the Federated Metadata Manager and Policy Enforcement Engine are critical for maintaining integrity and compliance, and the Distributed Compute Environment's Edge Node Cluster and Cloud Compute Nodes underscore the system's commitment to elastic scalability and efficient resource utilization.

[0052] The system further integrates an External Systems 121 interface via an API Gateway 114, which provides standardized, secure communication channels with external services. This interface enables seamless integration with external governance, control, and analytics applications, enhancing the overall utility of the ACDF system within broader data ecosystems.

[0053] Through this sophisticated architecture, the Enhanced ACDF System achieves unprecedented levels of security, adaptability, and operational efficiency. The interconnected modules collaboratively manage diverse data sources, dynamically optimize processing workflows, and ensure robust governance across distributed environments, thereby setting a new benchmark for modern, patentable data fabric systems.

FIG. 2A Flow chart Method for Dynamic data processing workflows in an Adaptive Cognitive Data Fabric

[0054] FIG. 2A is a flowchart that describes a method, according to some embodiments of the present disclosure. In some embodiments, at 230, the method may include continuously ingesting and reconciling heterogeneous data streams via a quantum-resistant data collector and self-healing reconciliation engine. At 231, the method may include analyzing the ingested data through a differential privacy-preserving semantic profiler to extract schema, content, and multi-modal semantic meaning. At 232, the method may include dynamically reconfiguring processing workflows using an adaptive pipeline orchestrator that leverages multi-agent reinforcement learning with a hierarchical reward mechanism and automatic feature engineering. At 233, the method may include triggering retraining of cognitive models through incremental knowledge distillation and curriculum-based prioritization upon detecting performance drift. At 234, the method may include maintaining immutable metadata and provenance records via a federated governance framework employing blockchain-based verification and zero-knowledge proofs.

FIG. 2 Method diagram of Advanced Cognitive Data Framework Methodology

FIG. 2 illustrates the comprehensive methodology for the Advanced Cognitive Data Framework (ACDF), which integrates diverse data sources, intelligent processing layers, adaptive workflow execution, cognitive model management, and rigorous governance controls. The system is architected to seamlessly manage the lifecycle of data—from ingestion and profiling to model intelligence and governance—with components designed for continuous adaptation and feedback.

[0055] The Multi-Source Data Ecosystem 210 is the foundational layer responsible for sourcing data from heterogeneous environments. It comprises three distinct repositories. The Structured Repositories 201 are implemented as databases optimized for well-defined, schema-based data. Streaming Sources 202, depicted as a cloud component, handle continuous, real-time data feeds that require rapid ingestion and processing. Unstructured Assets 203, represented by a node, manage data such as text, images, and multimedia that lack a predefined format. This diverse ecosystem ensures that the ACDF system can process a wide range of data types, setting the stage for subsequent intelligent processing.

[0056] The Cognitive Ingestion Layer 211 orchestrates the unified acquisition and initial preparation of raw data. Within this layer, the Dynamic ETL Orchestrator 204 automates the extraction, transformation, and loading of data, dynamically adapting to source-specific requirements and throughput variations. The Stream Processing Engine 205 processes continuous data flows, ensuring real-time handling and integration of streaming inputs. The Content Analysis System 206 performs preliminary content evaluation and classification, enabling downstream components to better contextualize and utilize the ingested data.

[0057] The Intelligent Profiling Engine 212 is dedicated to enriching the processed data with semantic context and monitoring changes. The Semantic Analyzer 207 evaluates data to generate meaningful representations, effectively translating raw inputs into structured insights. The Distribution Pattern Recognition 208 component examines data flow and distribution trends, identifying significant variations or anomalies. The Schema Evolution Detector 209 continuously monitors and detects changes in data structure, ensuring that the system remains responsive to modifications in source schemas or data characteristics. This layer plays a pivotal role in adapting the framework to dynamic data environments.

[0058] The Adaptive Pipeline Framework 223 is responsible for orchestrating the end-to-end execution of data processing workflows. At its core, the Reinforcement Learning Optimizer 213 utilizes feedback-driven algorithms to continuously refine and optimize workflow parameters. The Workflow Execution Mesh 214 manages the coordination of tasks across distributed nodes, ensuring seamless processing even in complex, multi-stage pipelines. The Pipeline Telemetry 215 component collects detailed execution metrics and performance data, which are essential for real-time monitoring and subsequent optimization of the processing pipeline.

[0059] The Model Intelligence Center 224 is focused on the cognitive aspects of the system, ensuring that analytical models remain accurate and responsive to data changes. The Cognitive Model Registry 216 serves as a repository for managing different versions of deployed models, ensuring proper version control and traceability. The Drift Detection Neural Network 217 monitors the performance of deployed models, identifying when changes in data patterns (concept drift) necessitate model updates. The Automated Retraining Orchestrator 218 responds to drift detection by initiating model retraining processes automatically, ensuring that the system adapts continuously to maintain high model accuracy and performance.

[0060] The Governance & Lineage Hub 225 enforces oversight, compliance, and traceability across the entire data processing lifecycle. The Provenance Graph Database 220 maintains a detailed, immutable record of data lineage, tracking the journey and transformations of data from ingestion to final output. The Metadata Knowledge Base 221 aggregates and organizes metadata associated with data assets, ensuring that context and data relationships are well-documented and accessible. The Governance Policy Engine 222 applies formalized rules and policies to the data processing workflows, ensuring that all operations comply with regulatory and organizational standards. This layer guarantees that every step within the ACDF framework is transparent, auditable, and secure.

[0061] In addition to the core functional flows, the ACDF methodology incorporates critical feedback loops and decision paths. The flow from the Multi-Source Data Ecosystem 210 to the Cognitive Ingestion Layer 211 signifies the initial intake of raw data, which is then transformed into a processed data stream delivered to the Intelligent Profiling Engine 212. The profiling output, enriched with semantic context and change metadata, informs the Adaptive Pipeline Framework, which in turn produces execution results that feed into the Model Intelligence Center 224. The model center monitors performance and triggers internal

retraining loops via drift detection 219, ensuring continuous cognitive adaptation. Finally, model versioning and documentation are seamlessly transmitted to the Governance & Lineage Hub 225, which, through its policy engine, provides feedback back to the ingestion layer for ongoing process refinement.

[0062] Emphasis is placed on components that introduce advanced adaptive capabilities and feedback mechanisms. The Intelligent Profiling Engine 212, through its integrated semantic analysis 207 and schema change detection 209, ensures that the framework remains agile in response to evolving data characteristics. The Adaptive Pipeline Framework's 223 reinforcement learning optimizer 213 and comprehensive telemetry 215 enable real-time adjustments to processing workflows. The Model Intelligence Center's 224 drift detection and automated retraining orchestrator 218 ensure that analytical models are perpetually optimized, reflecting the most current data trends. Finally, the Governance & Lineage Hub's 225 robust provenance tracking and policy enforcement mechanisms guarantee transparency and compliance throughout the data lifecycle.

[0063] The ACDF architectural methodology also includes a legend that defines the responsibilities of each layer, ensuring clarity in how source diversity management, unified data ingestion, semantic profiling, adaptive workflow execution, cognitive adaptation, and governance controls are implemented. This detailed, interlinked framework establishes the ACDF system as a state-of-the-art solution capable of intelligently managing and adapting to dynamic data environments with robust security and governance at its core.

FIG. 3 Method Diagram for Cognitive Intelligence Layer and Pipeline Execution Mesh

FIG. 3 illustrates a Method Diagram for Cognitive Intelligence Layer and Pipeline Execution Mesh, according to embodiments of the present disclosure. This architecture presents a robust and intelligent framework for managing distributed data pipelines, incorporating cognitive intelligence, real-time optimization, and adaptive orchestration mechanisms to ensure highly efficient and scalable data processing operations.

[0064] The Advanced Adaptive Pipeline Orchestration Architecture is structured into four core functional zones: the Data Acquisition Zone 307, the Cognitive Intelligence Layer 308, the Orchestration Control Plane 316, and the Pipeline Execution Mesh 317. Each of these zones encompasses specialized components that work synergistically to enable intelligent data ingestion, schema profiling, orchestration optimization, and dynamic execution management.

[0065] The Data Acquisition Zone 307 is responsible for sourcing and buffering multi-format data streams from heterogeneous sources. The Heterogeneous Data Sources 301 component serves as the entry point, accommodating diverse data formats originating from structured, semi-structured, and unstructured datasets. The Multi-Protocol Data Collector 302 provides a unified interface for ingesting data using multiple protocols, ensuring compatibility with a wide array of data sources. The Stream Buffering System 303 acts as an intermediary storage layer, temporarily holding raw data before further processing, ensuring lossless data ingestion while accommodating variable input rates.

[0066] The Cognitive Intelligence Layer 308 plays a crucial role in profiling and analyzing incoming data streams to extract semantic and structural insights. The Semantic & Structural Profiler 304 employs advanced pattern recognition techniques to derive schema and feature distributions dynamically. This profiler interfaces with the Semantic Schema Registry 305, a centralized repository that maintains metadata, schema definitions, and inferred data structures, enabling automated schema management. The Pattern Recognition Engine 306 applies time-series analysis and deep learning models to detect hidden patterns, correlations, and anomalies within ingested data, enhancing the system's ability to adaptively refine processing strategies.

[0067] The Orchestration Control Plane 316 serves as the decision-making and optimization hub of the architecture. The Adaptive Orchestration Controller 309 functions as the primary workflow manager, dynamically coordinating pipeline execution based on real-time insights from upstream components. This controller interacts with the Reinforcement Learning Optimizer 311, which leverages machine learning models to generate optimal workflow configurations, continuously refining execution strategies based on historical and real-time telemetry data. The Policy & Decision Cache 312 stores learned optimization strategies and precomputed execution policies, enabling rapid retrieval and deployment of optimized workflows. The Telemetry & Metric Collector 310 gathers performance statistics and operational feedback from all system components, ensuring data-driven decision-making in real-time pipeline execution.

[0068] The Pipeline Execution Mesh 317 is the execution backbone of the architecture, responsible for deploying, managing, and monitoring distributed pipeline processes. The Distributed Pipeline Executor 313 orchestrates execution across multiple computing nodes, ensuring parallelized and fault-tolerant operation. The Dynamic Resource Allocator 315 dynamically assigns computing resources based on real-time workload demands, optimizing

execution efficiency while minimizing resource wastage. The Execution Trace Repository 314 maintains detailed logs and execution traces, providing auditability, debugging insights, and performance tuning data. These execution logs are continuously analyzed by the Telemetry & Metric Collector to drive system-wide optimizations.

[0069] The architecture integrates multiple feedback loops to enable continuous adaptation and self-optimization. The Orchestration Control Plane 316 maintains a continuous cognitive feedback mechanism with the Cognitive Intelligence Layer 308, allowing real-time refinement of orchestration strategies based on evolving data characteristics. Similarly, the Execution Telemetry Loop ensures that the insights gathered from pipeline execution dynamically inform the orchestration decisions, creating a self-learning and self-adaptive system.

[0070] The Advanced Adaptive Pipeline Orchestration Architecture introduces several key innovations that enhance its efficiency and intelligence. The integration of Semantic-Aware Orchestration allows the system to infer schema structures dynamically, reducing manual configuration efforts and enhancing compatibility with diverse data formats. The Reinforcement Learning Optimization 311 ensures that workflow execution continuously improves over time by leveraging historical execution patterns and predictive analytics. The Real-time Pipeline Adaptation feature enables immediate response to data volume fluctuations, processing bottlenecks, and system failures, ensuring consistent pipeline efficiency. The Cognitive Feedback Loops create a self-reinforcing mechanism that enhances the system's adaptability, resilience, and overall intelligence.

[0071] Through this multi-layered architecture, the Advanced Adaptive Pipeline Orchestration Architecture achieves a novel paradigm in distributed data pipeline management. By combining intelligent data profiling, machine learning-driven decision-making, and dynamic resource allocation, this system optimizes pipeline performance, enhances scalability, and ensures resilience in complex data processing environments.

FIG. 4 Method Diagram for Multi-modal Semantic Intelligence Processing Fabric

FIG. 4 illustrates a Method Diagram for Multi-modal Semantic Intelligence Processing Fabric, which integrates heterogeneous data sources, cognitive intelligence, and adaptive orchestration for seamless semantic understanding across multiple modalities. This architecture provides a robust framework for ingesting, processing, fusing, and orchestrating multimodal data,

enabling cross-domain intelligence extraction with advanced contextual awareness and decision-making capabilities.

[0072] The Heterogeneous Data Ingestion Layer 415 serves as the primary entry point for diverse data types, consolidating structured, unstructured, visual, and IoT telemetry inputs into a unified processing pipeline. The Structured Data Gateway 402 manages relational and tabular data, ensuring schema alignment and consistency before semantic analysis. The Document Processing Pipeline 403 extracts contextual information from unstructured text, applying natural language processing techniques to enhance semantic representation. The Media Stream Receiver 405 ingests continuous audio, image, and video streams, segmenting and preprocessing raw data for downstream analysis. The IoT Telemetry Collector 404 normalizes real-time sensor and device signals, mapping telemetry data to standardized formats for correlation with other data sources. These components interface with the Data Source Federation 401, which acts as a synchronization hub to ensure consistency across different ingestion pathways, maintaining seamless interoperability among heterogeneous data formats.

[0073] The Multimodal Intelligence Processing Fabric 416 forms the core analytical layer, consisting of distinct subsystems specialized for different data modalities. The Natural Language Understanding (NLU) Layer 417 employs a Contextual Text Analyzer, which dissects linguistic structures and extracts semantic meanings. The Knowledge Extraction Engine 407 further distills relationships and key entities, storing derived representations within the Semantic Vector Store 408 for rapid retrieval and cross-modal fusion. These extracted knowledge elements interact with the visual intelligence system 418 to correlate text-based insights with image content, ensuring a richer semantic understanding.

[0074] The Visual Intelligence System 418 incorporates a Computer Vision Processor 409 that applies deep learning models to recognize objects, scenes, and spatial attributes. The Visual Context Recognition 410 module then refines these visual insights by associating detected elements with semantic labels, storing extracted features within the Visual Feature Repository 411. These features are referenced when aligning textual and acoustic insights, enabling a multimodal interpretation of content where images, text, and sound interact to produce a cohesive understanding.

[0075] The Acoustic Processing System 419 is responsible for deciphering sound-based information. The Audio Signal Analyzer 412 processes waveform and frequency patterns,

while the Speech & Sound Recognition 413 module transcribes speech, identifies background sounds, and classifies tonal variations. The extracted acoustic data is cataloged in the Acoustic Pattern Database 414, which is leveraged by the cross-modal fusion engine to refine speech-based sentiment detection, context derivation, and event recognition.

[0076] The Video Intelligence Framework 431 expands upon traditional image processing by incorporating temporal dynamics. The Spatiotemporal Analyzer 420 deconstructs video sequences, isolating motion trajectories, scene transitions, and action patterns. The Motion & Scene Interpreter 421 applies higher-level inference models to recognize behavioral patterns, interactions, and environmental cues, storing these interpretations in the Video Feature Timeline 422. This video-derived information enhances the cognitive fusion process, providing a temporal and spatial dimension to multimodal data integration.

[0077] The Cognitive Fusion & Semantic Integration Layer 432 unifies insights from diverse processing components. The Cross-Modal Fusion Engine 423 synthesizes vectors from text, image, audio, and video representations, ensuring cohesive multimodal knowledge representation. The Contextual Embedding Aligner 424 dynamically adjusts semantic embeddings to maintain contextual integrity across modalities, preventing loss of meaning due to data transformations. The Anomaly & Pattern Detector 425 continuously monitors fused data streams to identify inconsistencies, ensuring robust error correction and data validation. All these integrated semantic representations are stored within the Unified Semantic Graph 426, which acts as the central knowledge repository enabling complex cross-modal queries, inference derivation, and knowledge discovery.

[0078] The Intelligent Orchestration Plane 433 optimizes processing workflows, dynamically adjusting system parameters for efficiency and accuracy. The Adaptive Workflow Orchestrator 427 coordinates task execution, ensuring each subsystem operates with optimal resources. The Inferential Decision Engine 428 applies machine learning-driven heuristics to improve processing logic based on historical insights. The Cognitive Pipeline Controller 429 governs the execution sequence of multimodal tasks, adapting workflow structures based on evolving system feedback. The Orchestration Configuration Registry 430 maintains all system parameters and policies, providing a foundation for adaptive reconfiguration. This orchestration layer continuously refines its strategies through feedback loops, leveraging real-time performance metrics and semantic refinements to enhance overall intelligence operations.

[0079] Each layer within the architecture interacts through a structured set of feedback mechanisms and data flow pathways. The Semantic Refinement Signals ensure that outputs from the Cognitive Fusion & Semantic Integration Layer 432 inform the orchestration strategies, fine-tuning decision-making models over time. The Processing Optimization Parameters enable bidirectional coordination between the Intelligent Orchestration Plane 433 and the Multimodal Intelligence Processing Fabric 416, dynamically adjusting processing workloads to maximize efficiency. The Cross-Modal Context Enhancement allows the fusion layer to iteratively refine individual modality representations by drawing upon complementary insights from other domains. These interconnected pathways create a continuously evolving semantic intelligence system capable of adaptive learning and self-improving cognitive reasoning.

[0080] Through this layered, cross-modal integration, the Advanced Multimodal Semantic Intelligence Architecture achieves a comprehensive and scalable framework for deep semantic analysis, enabling robust and context-aware intelligence extraction across heterogeneous data sources. The architecture's adaptive orchestration, multimodal fusion, and inferential decision-making innovations provide a foundation for next-generation cognitive systems, transforming data interpretation across diverse domains such as healthcare, finance, surveillance, and automated assistance.

FIG. 5A Self-Adaptive Federated Governance and Policy Enforcement system

FIG. 5A is a block diagram that describes a self-adaptive Federated Governance and Policy Enforcement system 500, according to some embodiments of the present disclosure. In some embodiments, the self-adaptive adaptive Federated Governance and Policy Enforcement system 500 may include a central governance framework 527, intelligent metadata management 526, secure verification ecosystem 529, distributed autonomous nodes 528 components for orchestrating the Governance and policy enforcement according to some embodiments of the present disclosure.

[0081] The Federated Governance / Central Governance Framework 527 framework is designed to enforce decentralized governance policies across distributed data systems. This system allows multiple stakeholders to implement their own governance rules while ensuring consistency and compliance across the network. It utilizes a federated model that enables data stewardship to remain localized while providing centralized oversight for auditing and

compliance. The governance policies are dynamically applied across various nodes, ensuring that data handling and usage adhere to both local and global regulatory standards. The system integrates with external compliance tools and ensures the seamless enforcement of governance frameworks, promoting accountability and data security across the organization.

[0082] The Secure Verification Ecosystem 529 provides a robust mechanism for verifying data authenticity and integrity in a distributed environment. The ecosystem employs multiple layers of verification, including cryptographic signatures, consensus algorithms, and distributed ledger technologies, to ensure that all data transactions are secure and tamper-resistant. It integrates with the data pipeline to monitor and validate data at each stage, from ingestion to storage. This multi-tiered approach guarantees that only verified data is processed, significantly reducing the risks associated with data tampering and unauthorized access. The ecosystem also supports real-time verification of data, enabling the swift identification and rectification of discrepancies.

[0083] The Explainability Service 106 is a critical component designed to provide transparency and interpretability for complex machine learning models and algorithms. It employs advanced techniques, such as feature importance analysis, decision tree visualization, and model-agnostic interpretation methods, to provide human-readable explanations of model predictions. This service is integrated with both training and deployment environments, allowing users to inspect how models arrive at specific decisions and outcomes. By enhancing model transparency, the Explainability Service builds trust in AI systems and ensures that stakeholders can validate the logic behind automated decision-making processes, which is vital for compliance and user adoption.

[0084] The Intelligent Metadata Management 526 system is a central component for organizing and governing metadata across a distributed data ecosystem. This system captures detailed metadata, such as data lineage, access history, and transformation logs, providing a comprehensive view of data throughout its lifecycle. It employs machine learning techniques to automatically classify, tag, and organize metadata, improving the efficiency of data discovery and utilization. The system also integrates with data governance policies, ensuring that metadata is consistently updated, accurate, and aligned with regulatory requirements. By providing a unified metadata repository, this solution ensures better data visibility, traceability, and operational compliance.

[0085] The Carbon-Aware Scheduling 820 module optimizes resource allocation in computing environments by considering the environmental impact of resource usage. It incorporates real-time energy consumption data and carbon footprint analysis into the scheduling algorithm, dynamically adjusting workloads to minimize energy consumption during high-carbon periods. This approach ensures that computational tasks are completed in the most energy-efficient manner, reducing the system's overall carbon footprint. The Carbon-Aware Scheduling system can be integrated with cloud providers, edge devices, and on-premise infrastructures to promote sustainable computing practices, aligning with green initiatives and reducing operational costs associated with energy consumption.

[0086] The Distributed Autonomous Nodes 528 system is designed to enable a network of self-governing nodes that operate without centralized control. Each node in this system is capable of making independent decisions based on predefined rules and environmental factors. The nodes communicate and collaborate with each other to achieve consensus and optimize overall system performance. By leveraging machine learning and distributed algorithms, the system ensures that nodes can dynamically adapt to changes in their environment, making real-time adjustments to maintain operational efficiency. This architecture promotes scalability, fault tolerance, and decentralization, making it ideal for large-scale distributed systems.

[0087] The Anomaly Detection Engine 517 is a specialized module that continuously monitors data streams to identify abnormal patterns and outliers. It utilizes machine learning models, such as neural networks and clustering algorithms, to detect both known and unknown anomalies in real time. The engine applies statistical analysis and temporal modeling techniques to recognize patterns that deviate from expected behaviors, enabling the early detection of potential issues such as fraud, data corruption, or system malfunctions. By leveraging historical data and adaptive learning algorithms, the Anomaly Detection Engine can continuously improve its accuracy and reliability in identifying anomalies, providing real-time alerts and insights to stakeholders.

[0088] The Quantum-Resistant Cryptographic 523 Layer is a security framework designed to protect data from the potential threats posed by quantum computing. This layer employs post-quantum cryptography algorithms that are resistant to attacks by quantum computers, ensuring that encrypted data remains secure even as quantum technologies evolve. The system integrates these advanced cryptographic techniques with existing security protocols, including encryption, digital signatures, and secure key exchange methods, to provide a multi-layered

defense against unauthorized access and data breaches. By proactively adopting quantum-resistant methods, this layer future-proofs the security infrastructure, ensuring that sensitive data remains protected in a quantum-computing world.

FIG. 5 Advanced Adaptive Federated Metadata Management System with Distributed Consensus - System Architecture

FIG. 5 illustrates the architecture diagram of an Advanced Adaptive Federated Metadata Management System with Distributed Consensus, according to embodiments of the present disclosure. This system establishes a scalable and intelligent framework for federated metadata governance across distributed environments, incorporating cryptographic verification, anomaly detection, and dynamic policy orchestration to ensure compliance and consistency across multiple autonomous nodes.

[0089] The Advanced Adaptive Federated Metadata Management System comprises four core subsystems: the Central Governance Framework 527, the Distributed Autonomous Nodes 528, the Secure Verification Ecosystem 529, and the Intelligent Metadata Management module. These components work in tandem to establish a distributed yet coordinated approach to metadata governance, ensuring real-time compliance, security, and adaptability. Each module contains specialized functionalities that enable decentralized metadata management while maintaining consistency through a distributed consensus mechanism.

[0090] The Central Governance Framework 527 serves as the authoritative control center, comprising several key components. The Central Metadata Authority 506 is responsible for defining and maintaining metadata governance policies and ensuring synchronization across the distributed network. It interfaces directly with the Adaptive Policy Orchestration Engine 511, which dynamically adjusts governance rules based on compliance feedback, usage patterns, and anomaly reports. The Conflict Resolution System 505 acts as a dispute resolution mechanism that reconciles conflicting metadata definitions between autonomous nodes. The Taxonomy Versioning Controller 507 maintains schema evolution and ensures seamless updates across hierarchical nodes. The Metadata Lineage Tracker 508 records provenance information for metadata changes, ensuring auditability and transparency. All governance-related data is stored in the Holistic Governance Repository 509, which acts as the central repository for metadata policies, compliance records, and version-controlled metadata schemas. The Real-Time Compliance Monitor 510 continuously assesses policy adherence,

providing feedback to the Adaptive Policy Orchestration Engine for necessary adjustments. The Central Metadata Authority 506, through its components, establishes authoritative metadata definitions and continuously synchronizes updates across the federated system, ensuring adherence to dynamically evolving governance policies.

[0091] [0071] The Distributed Autonomous Nodes 528 constitute the decentralized metadata governance layer, where metadata is managed locally at hierarchical levels while ensuring compliance with central policies. Each hierarchical node (512, 513, 518) operates independently but follows the global metadata governance policies dictated by the Central Metadata Authority. These nodes interface with localized metadata caches (514, 515, 519) to store frequently accessed metadata and minimize synchronization overhead. Additionally, each node is equipped with an Anomaly Detection Engine (516, 517, 520), which monitors metadata integrity and detects inconsistencies before they propagate across the system. The anomaly detection engines (516, 517, 520) continuously scan metadata for deviations from established governance rules and notify the Central Metadata Authority of potential conflicts, ensuring early identification and resolution of inconsistencies. Each hierarchical node (512, 513, 518) securely distributes metadata updates via the Distributed Ledger Coordinator, which enforces consensus and prevents unauthorized modifications, ensuring synchronized metadata governance across the federated infrastructure.

[0092] The Secure Verification Ecosystem 529 introduces cryptographic mechanisms to ensure data integrity, privacy, and regulatory compliance. The Distributed Ledger Coordinator 521 plays a critical role in registering immutable metadata references, ensuring that all metadata updates are verifiable and tamper-resistant. It interfaces with the Zero-Knowledge Proof Generator 522 to generate cryptographic proofs for access control and metadata validation. These proofs are processed through the Quantum-Resistant Cryptographic Layer 523, which applies post-quantum encryption standards to protect against both classical and quantum computing threats. The Smart Contract Executor 525 automates policy enforcement by executing predefined governance rules upon metadata updates. The Temporal Metadata Validator 524 ensures time-based consistency by validating metadata updates against historical records before allowing integration into the distributed ledger. The interaction between these components ensures that all metadata transactions are secure, compliant, and resistant to unauthorized modifications, providing an immutable audit trail that maintains governance integrity.

[0093] The Intelligent Metadata Management 526 module provides advanced processing capabilities for metadata classification, reasoning, and quality assurance. The Metadata Classification Service 501 is responsible for categorizing metadata based on predefined taxonomies and contextual attributes. It interacts with the Ontology Inference Engine 502, which derives semantic relationships and hierarchical dependencies among metadata entities. The Semantic Reasoner 503 applies logic-based inference to establish contextual meaning, enabling cross-domain metadata integration. The Data Quality Assurance Framework 504 evaluates metadata consistency, accuracy, and reliability using automated validation techniques. If conflicting metadata definitions are detected, the Conflict Resolution System 505 intervenes by proposing reconciliation strategies based on predefined policies. This intelligent processing pipeline ensures that metadata is not only structured and organized but also contextually enriched and validated before integration into the federated metadata governance framework.

[0094] The system's operational workflow ensures end-to-end metadata governance across all layers. The Central Governance Framework 527 publishes authoritative metadata definitions to the Holistic Governance Repository 509, which then distributes policies to the Distributed Autonomous Nodes. Each node validates metadata against compliance policies using its anomaly detection engines (516, 517, 520), before storing validated records in the local metadata caches (514, 515, 519). Secure verification mechanisms further enhance governance by ensuring that all metadata transactions are cryptographically protected and compliant with established policies. The intelligent metadata processing pipeline continuously classifies, validates, and refines metadata definitions, ensuring high-quality, semantically consistent metadata management across the distributed ecosystem.

[0095] Through its distributed and adaptive architecture, the Advanced Adaptive Federated Metadata Management System ensures scalable, secure, and context-aware metadata governance. The integration of cryptographic verification, anomaly detection, and intelligent classification enables autonomous nodes to maintain metadata consistency while adapting to evolving governance requirements. The system's federated approach ensures that metadata policies are enforced dynamically across distributed environments while preserving decentralized control, making it suitable for regulatory-compliant and large-scale metadata management applications.

FIG. 6 Adaptive Cognitive Data Fabric - Federated Data Governance and Cognitive Model Management

FIG. 6 illustrates the method diagram of Adaptive Cognitive Data Fabric - Federated Data Governance and Cognitive Model Management, according to embodiments of the present disclosure. This system provides an intelligent, self-optimizing framework for data ingestion, processing, governance, and resource optimization, incorporating cognitive modeling techniques and federated governance mechanisms to ensure efficient and secure data management.

[0096] The Federated Data Governance and Cognitive Model Management method is composed of five primary sections: Data Ingestion & Profiling, Adaptive Pipeline Orchestration, Cognitive Model Management, Federated Data Governance, and Resource Optimization. These components work cohesively to enable end-to-end automation, real-time governance enforcement, and dynamic resource allocation based on workload and environmental factors.

[0097] The Data Ingestion & Profiling 821 section is responsible for acquiring and analyzing data from multiple sources. The Data Collector 801 ingests both structured and unstructured data from diverse origins, ensuring broad compatibility with external data sources. The Stream Reconciliation Engine 802 ensures data consistency and Byzantine fault tolerance by detecting and resolving inconsistencies between ingested data streams. The Real-Time Semantic Profiler 803 applies natural language processing and computer vision techniques to extract schema definitions, content structure, and semantic meaning from incoming data, enriching the dataset with intelligent profiling metadata. Once data is reconciled, it is forwarded to the Adaptive Pipeline Orchestrator 822 for further processing. The profiling metadata is also logged into the Centralized Metadata Repository for governance purposes.

[0098] The Adaptive Pipeline Orchestration 822 section dynamically configures and manages data processing pipelines using reinforcement learning techniques. The Adaptive Pipeline Orchestrator 804 employs a multi-agent reinforcement learning framework with a hierarchical reward system to optimize pipeline configurations based on data characteristics and workload demands. It triggers the Automatic Feature Engineering module 805, which creates and selects optimal feature transformations based on the semantic characteristics of the data. Adaptive Pipeline Orchestration 822 also interacts with the Performance Monitoring

Module 812 in the Cognitive Model Management 824 section to fine-tune pipeline operations based on model output performance and user feedback.

[0099] The Cognitive Model Management 824 section ensures the reliability and adaptability of deployed models. The Performance Monitoring Module 812 continuously tracks model outputs against ground truth values and user feedback, sending performance metrics to the Drift Detection Module 813 and it employs multivariate statistical process control techniques to detect distributional shifts in input data, triggering the Automated Retraining Orchestrator when significant drift is detected. Automated Retraining Orchestrator 814 employs curriculum learning and transfer learning strategies to retrain models efficiently. The Model Versioning System 815 maintains a directed acyclic graph (DAG) structure to manage model lineage, allowing seamless rollback and version tracking. The latest model versions are then provided to Performance Monitoring Module 812 for continuous monitoring.

[0100] The Federated Data Governance 823 section enforces policy compliance, ensures auditability, and maintains metadata integrity. The Centralized Metadata Repository 807 serves as the authoritative source for metadata, logging information from ingestion, pipeline operations, and model retraining events. Blockchain-based Verification 806 ensures the immutability of recorded metadata by cryptographically securing governance records. The Policy Enforcement Engine 808 translates high-level governance policies into implementation rules, ensuring compliance across all system components. The Real-Time Access Control 809 module enforces security using homomorphic encryption techniques to protect sensitive data while enabling computations on encrypted datasets. The Automated Lineage Tracker 810 records all data transformations and model interactions, creating a comprehensive audit trail. The Cross-Environment Synchronization 811 module ensures metadata consistency across cloud and on-premises environments, preventing governance discrepancies.

[0101] The Resource Optimization section enables efficient allocation and utilization of computational resources. The Workload Analyzer 816 profiles computational demands using hardware-specific models, providing insights into optimal resource distribution. Based on this analysis, the Predictive Resource Allocator 817 implements a market-based bidding system to allocate computing resources dynamically. The Dynamic Scaling Engine 818 provisions and de-provisions resources in real-time, ensuring cost-effective scaling. The Cost Optimization Module 819 balances system performance against resource expenditure, reducing operational

overhead. The Energy Efficiency Controller 820 employs carbon-aware scheduling algorithms to optimize energy consumption, contributing to sustainable computing practices.

[0102] The data flow between sections follows a structured pipeline. The Data Collector 801 ingests raw data, which is reconciled by the Stream Reconciliation Engine before being analyzed by the Real-Time Semantic Profiler. Profiling results are then forwarded to the Adaptive Pipeline Orchestrator, which configures pipeline execution. Feature transformations are optimized by the Automatic Feature Engineering module before the data enters cognitive modeling workflows. Performance metrics from model executions are monitored and fed back into the orchestration layer, ensuring continuous improvements.

[0103] The governance flow ensures metadata integrity and compliance enforcement. The Real-Time Semantic Profiler 803, Adaptive Pipeline Orchestrator 804, and Automated Retraining Orchestrator log metadata into the Centralized Metadata Repository 807. Governance rules are enforced through the Policy Enforcement Engine 808, which manages access control via the Real-Time Access Control 809 module. Blockchain-based Verification 809 guarantees metadata authenticity, while the Automated Lineage Tracker records all data transformations. The Cross-Environment Synchronization module maintains metadata consistency across distributed deployments.

[0104] The resource optimization flow dynamically allocates computational resources based on workload demand. The Workload Analyzer 816 analyzes system performance and sends workload patterns to the Predictive Resource Allocator 817 and it dynamically provisions resources through the Dynamic Scaling Engine 818, which then optimizes resource allocation through the Cost Optimization Module 819. Energy efficiency 820 considerations are incorporated into the allocation strategy using feedback from the Energy Efficiency Controller.

[0105] The feedback loops within the ACDF architecture enable continuous adaptation and self-optimization. The Performance Monitoring Module 812 provides feedback to the Adaptive Pipeline Orchestrator 804 regarding data quality and model performance. The Automated Retraining Orchestrator informs Adaptive Pipeline Orchestrator about retraining outcomes, ensuring that future pipeline configurations consider historical training performance. The Energy Efficiency Controller supplies carbon impact and energy metrics to the Predictive Resource Allocator, refining resource allocation strategies based on sustainability goals.

[0106] Through this architecture, the Adaptive Cognitive Data Fabric achieves intelligent, scalable, and secure data processing. The system autonomously adapts to evolving data landscapes, optimizes resource allocation based on computational demand, and ensures comprehensive governance enforcement through federated policies. By leveraging advanced reinforcement learning, cognitive modeling, and blockchain-based verification 806, Adaptive Cognitive Data Fabric establishes a resilient and self-optimizing framework for next-generation data ecosystems.

[0107] FIG. 3 is a block diagram that describes a self-adaptive federated governance and policy enforcement system 300, according to some embodiments of the present disclosure. In some embodiments, the self-adaptive federated governance and policy enforcement system 300 may include a federated metadata manager 310 that maintains a centralized yet distributed repository of data catalogs with blockchain-based immutable tracking and zero-knowledge proofs and an explainability service 330 that generates counterfactual explanations through causal model extraction to enhance decision transparency.

[0108] In some embodiments, the self-adaptive federated governance and policy enforcement system 300 may also include a policy enforcement engine 320 that translates natural language policies into formally verified, rule-based implementations and utilizes homomorphic encryption for real-time access control. The self-adaptive federated governance and policy enforcement system 300 may also include a resource optimization module 340 integrating a market-based predictive resource allocator, dynamic scaling engine, topology-aware peer discovery for secure P2P mesh communication, and carbon-aware scheduling to optimize computing resources across heterogeneous environments.

FIG. 7 System and Method for Dependent Claims Detailed Description

Adaptive Pipeline Orchestrator with Multi-Agent Reinforcement Learning 710

[0109] The Adaptive Pipeline Orchestrator utilizes a Multi-Agent Reinforcement Learning 710 framework to continuously optimize workflow execution across distributed computational environments. It dynamically adjusts workflow configurations based on historical performance metrics, real-time feedback, and computational constraints, ensuring maximized efficiency and resource utilization. The orchestrator employs a hierarchical reward system that balances immediate processing efficiency with long-term learning objectives. It enables intelligent workflow adaptation by integrating predictive telemetry, automated model retraining triggers,

and dynamic load balancing, ensuring adaptive response to evolving data and system conditions.

Real-Time Semantic Profiler with Multi-Modal Vector Embedding Alignment

[0110] The Real-Time Semantic Profiler 720 implements an advanced multi-modal vector embedding alignment technique to maintain a unified semantic representation across diverse data types. It dynamically adjusts embedding dimensions based on real-time semantic similarity metrics, ensuring optimized feature space consistency for structured, unstructured, and time-series data. The system integrates differential privacy mechanisms, preserving confidentiality while maintaining high-fidelity embeddings. By leveraging adaptive dimensionality reduction and self-supervised learning, it mitigates representational drift and enhances real-time inferencing capabilities, ensuring robust and contextually relevant semantic analysis across large-scale data streams.

Cognitive Model Re-trainer with Curriculum Learning and Directed Acyclic Graph Model Derivation

[0111] The Cognitive Model Re-trainer 730 utilizes curriculum learning to systematically refine model performance by prioritizing retraining on data samples that address critical performance gaps. It maintains a Directed Acyclic Graph structure to track model lineage, enabling automated version comparison, rollback, and performance validation. The system incorporates real-time drift detection, dynamically identifying evolving data patterns and triggering targeted updates. By integrating incremental knowledge distillation and continual learning guarantees, it ensures adaptive model evolution, minimizing catastrophic forgetting while maintaining high generalization across dynamic and multi-modal data environments.

Explainability Service with Counterfactual Explanation Generation

[0112] The Explainability Service 740 generates precise counterfactual explanations by utilizing an invertible transformation map that establishes a bidirectional traceability pathway between input features and model decisions. The system enables real-time interpretability by extracting causal relationships, allowing users to understand the rationale behind model predictions. It integrates an advanced causal inference engine that computes alternative decision scenarios, identifying key factors influencing outcomes. By offering an interactive

and mathematically validated explanation framework, it enhances regulatory compliance, risk assessment, and user trust in artificial intelligence-driven decision-making.

Federated Metadata Manager with Blockchain-Based Compliance Verification

[0113] The Federated Metadata Manager 750 enforces cross-jurisdictional compliance verification by integrating an immutable blockchain-based provenance tracking system with selective disclosure protocols. It ensures data integrity, regulatory adherence, and governance enforcement across distributed environments. The system utilizes zero-knowledge proofs to cryptographically validate metadata access without exposing sensitive information. By leveraging smart contract-based compliance automation, it dynamically aligns governance rules with evolving regulations. The metadata manager enables tamper-proof policy enforcement, decentralized audit trails, and intelligent access controls, ensuring regulatory accountability while preserving operational transparency across multi-entity federated systems.

Topology-Aware Peer Discovery for Distributed Compute Optimization

[0114] The Topology-Aware Peer Discovery System 760 orchestrates a self-organizing compute mesh that dynamically redistributes workloads based on real-time network conditions, hardware capabilities, and energy efficiency metrics. It integrates predictive modeling to anticipate network congestion and computational resource availability, enabling proactive load balancing across cloud, edge, and on-premise nodes. The system ensures low-latency compute offloading by employing intelligent peer-to-peer workload orchestration. By continuously adapting to system topology changes, it optimizes distributed execution efficiency, minimizes latency bottlenecks, and enhances overall computational resilience in large-scale heterogeneous processing environments.

Quantum-Resistant Data Collector with Post-Quantum Cryptographic Security

[0115] The Quantum-Resistant Data Collector 770 secures data transmission using forward secrecy protocols integrated with post-quantum cryptographic primitives to prevent vulnerabilities against quantum computing threats. It employs lattice-based encryption, hash-based signatures, and dynamic key rotation techniques to maintain long-term confidentiality. The system incorporates secure multiparty computation, enabling encrypted data processing across multiple trust zones without exposing raw information. By integrating quantum-resistant authentication and zero-trust principles, it ensures robust security, protecting sensitive data

assets against evolving cryptographic threats while maintaining high-performance data ingestion and processing integrity.

Natural Language Policy Compiler for Governance Enforcement

[0116] The Natural Language Policy Compiler 780 translates high-level governance policies into formally verified, mathematically provable enforcement mechanisms across diverse technology stacks. It utilizes logical inference models, rule-based reasoning engines, and cryptographic attestation to ensure policy execution integrity. The system integrates blockchain-backed compliance verification, providing immutable regulatory enforcement transparency. By dynamically adapting to regulatory changes, it ensures enterprise-wide policy consistency, automated risk assessment, and real-time governance enforcement. The compiler's explainable rule transformation framework guarantees seamless integration across cloud, hybrid, and on-premise infrastructures while maintaining legal and operational compliance.

I/WE CLAIM

INDEPENDENT CLAIMS

Claim 1: A self-evolving adaptive cognitive data fabric system comprising:

[0117] a quantum-resistant data collector configured for secure, multi-protocol ingestion with dynamic key rotation and post-quantum cryptographic algorithms;

[0118] a stream reconciliation engine that implements self-healing consistency with Byzantine fault tolerance to reliably synchronize heterogeneous data streams;

[0119] a real-time semantic profiler that utilizes differential privacy-preserving vector embedding generation by integrating natural language processing and computer vision for multi-modal data analysis;

[0120] an adaptive pipeline orchestrator employing a multi-agent reinforcement learning framework with a hierarchical reward system and automatic feature engineering based on semantic feedback;

[0121] a cognitive model re-trainer configured for incremental knowledge distillation with continual learning and curriculum-based retraining to address concept drift; and

[0122] a federated metadata manager that enforces immutable, blockchain-based provenance tracking with zero-knowledge proofs.

Claim 2: A method for dynamic optimization of data processing workflows in an adaptive cognitive data fabric, comprising:

[0123] continuously ingesting and reconciling heterogeneous data streams via a quantum-resistant data collector and self-healing reconciliation engine;

[0124] analyzing the ingested data through a differential privacy-preserving semantic profiler to extract schema, content, and multi-modal semantic meaning;

[0125] dynamically reconfiguring processing workflows using an adaptive pipeline orchestrator that leverages multi-agent reinforcement learning with a hierarchical reward mechanism and automatic feature engineering;

[0126] triggering retraining of cognitive models through incremental knowledge distillation and curriculum-based prioritization upon detecting performance drift; and

[0127] maintaining immutable metadata and provenance records via a federated governance framework employing blockchain-based verification and zero-knowledge proofs.

Claim 3: A self-adaptive federated governance and policy enforcement system for a cognitive data fabric, comprising:

[0128] a federated metadata manager that maintains a centralized yet distributed repository of data catalogs with blockchain-based immutable tracking and zero-knowledge proofs;

[0129] a policy enforcement engine that translates natural language policies into formally verified, rule-based implementations and utilizes homomorphic encryption for real-time access control;

[0130] an explainability service that generates counterfactual explanations through causal model extraction to enhance decision transparency; and

[0131] a resource optimization module integrating a market-based predictive resource allocator, dynamic scaling engine, topology-aware peer discovery for secure P2P mesh communication, and carbon-aware scheduling to optimize computing resources across heterogeneous environments.

DEPENDENT CLAIMS:

[0132] The system of Final Independent Claim 1, wherein the adaptive pipeline orchestrator employs a multi-agent reinforcement learning framework with a hierarchical reward system balancing immediate processing efficiency, computational constraints, and long-term learning objectives.

[0133] The system of Final Independent Claim 2, wherein the real-time semantic profiler implements multi-modal vector embedding alignment that dynamically adjusts embedding dimensions based on semantic similarity metrics to maintain a unified representation across diverse data types.

[0134] The system of Final Independent Claim 1, wherein the cognitive model re-trainer implements curriculum learning to prioritize retraining on samples that most effectively

address performance gaps while maintaining a directed acyclic graph (DAG) of model derivations for automated performance comparison.

[0135] The system of Final Independent Claim 3, wherein the explainability service generates counterfactual explanations using an invertible transformation map between input features and model decisions to enable bidirectional traceability of decision pathways.

[0136] The system of Final Independent Claim 1, wherein the federated metadata manager implements cross-jurisdictional compliance verification using selective disclosure protocols integrated with blockchain-based immutable record-keeping.

[0137] The system of Final Independent Claim 3, wherein the topology-aware peer discovery system implements a self-organizing compute mesh that dynamically redistributes workloads based on network conditions, hardware capabilities, and energy efficiency metrics.

[0138] The system of Final Independent Claim 1, wherein the quantum-resistant data collector incorporates forward secrecy protocols with post-quantum cryptographic primitives to ensure robust security against future quantum computing attacks.

[0139] The system of Final Independent Claim 3, wherein the natural language policy compiler automatically translates high-level governance policies into formally verified, rule-based enforcement mechanisms with mathematical guarantees of compliance across diverse technology stacks.

ABSTRACT

[0140] A self-evolving adaptive cognitive data fabric system with federated governance, comprising: a quantum-resistant data collector for secure, multi-protocol ingestion with dynamic key rotation and post-quantum cryptographic algorithms; a self-healing stream reconciliation engine with Byzantine fault tolerance to ensure reliable synchronization of heterogeneous data streams; a real-time semantic profiler integrating natural language processing and computer vision for multi-modal data analysis while preserving differential privacy; an adaptive pipeline orchestrator employing multi-agent reinforcement learning for dynamic optimization of data workflows with hierarchical reward mechanisms; a cognitive model re-trainer leveraging continual learning and curriculum-based knowledge distillation to counter concept drift; and a federated metadata manager enforcing immutable, blockchain-based provenance tracking with zero-knowledge proofs. The system dynamically reconfigures processing workflows, optimizes AI model training across distributed environments, and ensures compliance with regulatory policies while enabling intelligent, scalable, and resilient data management.